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Novel Single-Point-Type Tactile Pressure Sensors for Earth and Soil-Structure Interface Contact Pressures Measurement

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Reference

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ABSTRACT

The single-point-type tactile pressure sensor (TPS) is a thin and flexible piezoresistive sensor that shows promise in overcoming the limitations of current methodologies for measuring earth pressure within soils and contact pressure at soil-structure interfaces. Despite this potential, the feasibility of the TPS application has not been verified in physical model tests of soil-structure interaction. This article addresses this gap by introducing a new earth pressure cell (TPS-EPC) with limited thickness and a soil-structure interface contact pressure transducer (TPS-ICPT) conforming to curved interfaces based on a single-point-type TPS. The article presents a comprehensive methodology encompassing sensor design, signal processing, and calibration. Additionally, a nonlinear signal conversion method is recommended to enhance measurement accuracy. Compared with traditional EPCs, TPS-EPCs with the suggested methodology offer more precise earth pressure measurements due to reduced arching effects resulting from their limited thickness. The feasibility of new sensors was validated for the first time in the physical modeling of cyclic axial soil pipe interaction, providing a new and highly sensitive method for soil pressure measurement and yielding valuable insights into soil-pipe interaction behavior, including constrained dilation on the soil-pipe interface and the evolution of soil arching on the pipe.

Keywords

single-point-type tactile pressure sensors, earth pressure cell, soil-structure interface contact pressure transducer, sensor accuracy, soil-structure interaction, physical modeling

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Introduction

Tactile pressure sensor (TPS), also known as film-like pressure sensor, is a type of polymer piezoresistive sensor. Its electrical conductance increases with the application of a higher pressure. It could be used to monitor the changes in earth pressure within the soil and contact pressure at the soil-structure interface, which are crucial for understanding the interaction between soil and buried structure. The thickness of traditional earth pressure cells (EPCs) (usually more than 5 mm), coupled with their higher stiffness, may cause soil arching around the sensor, normally leading to an overestimation of earth pressure (Tory and Sparrow 1967; Labuz and Theroux 2005). With a smaller thickness of no more than 0.5 mm, TPS is expected to provide more reliable earth pressure measurement. It is also flexible, allowing it to conform to the curved surfaces of common geotechnical structures like piles and pipes, compared with traditional soil-structure interface contact pressure transducers (ICPTs).

TPS can be categorized into two main types based on the number of measuring points: array-type and single-point-type. The array-type TPS typically comprises a grid matrix of small sensing points with the ability to detect hundreds of thousands of measuring points (Weidlich and Achmus 2008; Palmer et al. 2009; Gillis, Dashti, and Hashash 2015; Muszynski et al. 2016; Madabhushi and Haigh 2019; Meguid and Ahmed 2020). The single-point-type TPS, on the other hand, has only one sensing area with a diameter or size length ranging from 3 to 80 mm (Interlink Electronics 2023; Tekscan 2023). The current study focuses on the single-point-type TPS since it can be used with a common data logger that is more affordable.

However, the current use of single-point-type TPS mainly focuses on the measurement of contact pressure at flat interfaces in laboratory tests. Kheiripour Langroudi et al. (2010) pasted a single-point TPS on an interface direct shear box to measure the spatial nonuniformities and temporal fluctuations of normal pressure during the interface shear. Gao and Wang (2014) used this sensor to measure lateral earth pressure in one-dimensional consolidation testing using an oedometer. The single-point TPS was also developed to measure excess pore water pressure during consolidation using a specialized needle probe. Kootahi and Leung (2022) conducted a study on the size effects of EPC sensing area using the single-point-type TPS in an oedometer. There have been no studies applying the single-point-type TPS in large-scale physical modeling tests or field monitoring. Its feasibility to measure the earth pressure and contact pressure at curved interfaces has not been confirmed. Furthermore, the input-output behavior of TPS exhibits significant nonlinearity and hysteresis and is influenced by specific installation conditions (Paikowsky and Hajduk 1997; Muszynski et al. 2016; Liu, Xu, and Zhang 2021; Kootahi and Leung 2023). It is necessary to establish a reliable methodology for using the single-point-type TPS that can measure pressures with greater accuracy.

This study introduces a new EPC (TPS-EPC) and a new soil-structure ICPT (TPS-ICPT), based on the single-point-type TPS, and a corresponding methodology for measuring earth pressure in soils and contact pressures at curved interfaces, respectively. Subsequently, the performance of the TPS-EPC, using the proposed methodology, is compared with that of the traditional EPC in an oedometer. Finally, the feasibility of TPS-based sensors is validated through axial soil-pipe interaction physical modeling.

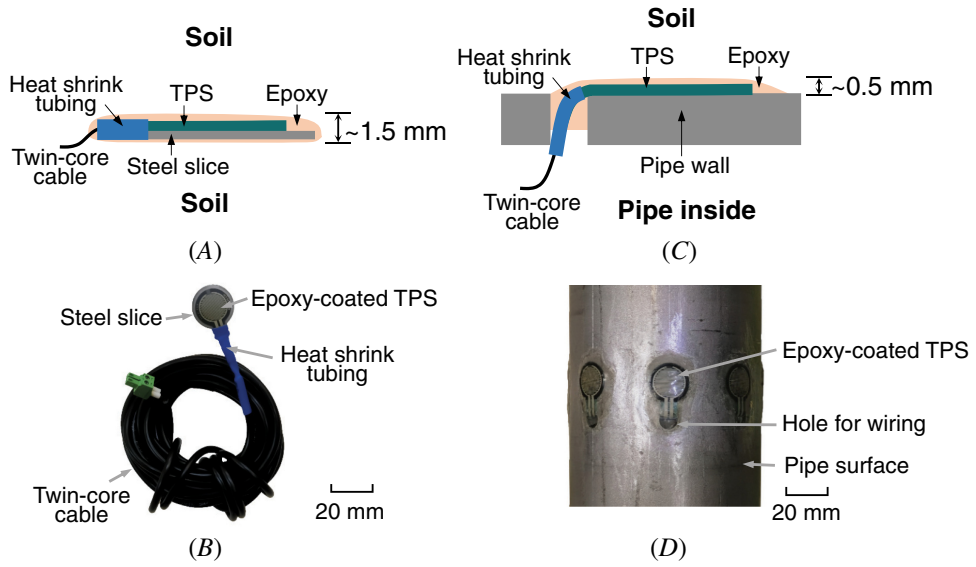
Design of Thin TPS-EPS and Curved Surface-Fitted TPS-ICPT

A typical single-point-type TPS manufactured by Interlink Electronics Force Sensing Resistor (FSR) 402 was used. The active area has a diameter of 12.7 mm, which is larger than 30 d_{50} of the sand used in this study, sufficiently avoiding the particle size effects on the measurement accuracy based on the results of Kootahi and Leung (2022). The nominal thickness of the sensor is 0.46 mm, no more than 1.25 d_{50} of the sand used in this study. More details can be found in Interlink Electronics (2023).

Two new transducers (i.e., TPS-EPC and TPS-ICPT) were developed. **Figure 1*A*** and **1*B*** show a schematic diagram and photograph of the TPS-EPC. TPS could not be buried in the soil directly because the soil deformation may stretch or bend the flexible sensors, resulting in unpredictable effects on the TPS's sensitivity and zero offset (Interlink Electronics 2023). Therefore, TPS is stuck on a 0.5-mm-thickness insulating-paint–coated steel slice

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FIG. 1 Design of TPS-based sensors: (*A*) schematic side view of TPS-based EPC; (*B*) photograph of TPS-based EPC; (*C*) schematic side view of TPS-based ICPT; (*D*) photograph of TPS-based ICPT.



whose shape adapts to TPS via water-cutting. Meanwhile, a coating layer by Epoxy Adhesive DP100 manufactured by 3M is used to protect the TPS. After 48 h of curing, the epoxy adhesive would form a stiff coating layer with a thickness of around 0.5 mm. The coating may affect the sensitivity of TPS, and 18 coated TPS-based sensors were calibrated with a loading path of 0∼60 kPa and a pressure loading rate of 4 kPa/min. Their average sensitivity is 7.75 $(\text{GPa}\cdot\Omega)^{-1}$, which is sufficiently sensitive for measuring earth and soil-structure interface pressures. More details of calibration are discussed later. The short tail version of FSR 402 (FSR 402 Short) is used in TPS-EPC for a smaller volume. A heat shrink tube is set at the tail of TPS and steel slice to further protect the wiring position. The outsides of the slice's tail have a couple of grooves for better fixing the heat shrink tube. The thickness of the new sensor is around 1.5 mm, equaling 4 d_{50} of the sand used in this study, significantly smaller than the thickness of the commonly used EPC. The effects of soil arching on TPS-EPC are verified and compared with traditional EPC later.

The schematic diagram and photograph of the TPS-ICPT on a pipe surface are illustrated in **figure 1C** and **1D**. The long-tail version of FSR 402 is used in TPS-ICPT for better wiring. Holes with a diameter of 8 mm are opened on the pipe. After wiring through the holes, the FSR 402 is stuck on the pipe surface by making use of its flexibility and coated using the same method in TPS-EPC. The total thickness of FSR 402 and the epoxy coating is around 0.5 mm, close to d_{50} of the sand used in this study. Hence, the effects of thickness are negligible.

Based on the results of some trial tests, the threshold pressure that induces an obvious change in the electrical conductance (>1 MPa^{-1}) of bare FSR 402 varies from 0∼10 kPa. Increasing the surface roughness of the pipe or steel slice was found to be able to reduce the threshold pressure to almost zero in all cases. The working principle of this method is probably due to the more significant prestressing during the sticking process for a rougher surface. The surface roughness can be controlled using mechanical surface treatment technology directly or mixing some silty soil evenly into the epoxy adhesive used between TPS and the surface of the pipe or slice.

Multichannel Measurement

The output signal of single-point-type TPS is its electrical conductance (G) [the reciprocal of electrical resistance (R)]. It cannot be directly measured by common data loggers used in geotechnical testing.

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FIG. 2 Measurement wiring and method: (*A*) method I: resistance-voltage conversion by electrical interface; (*B*) method II: 4-wire electrical resistance measurement by dataTaker DT85.

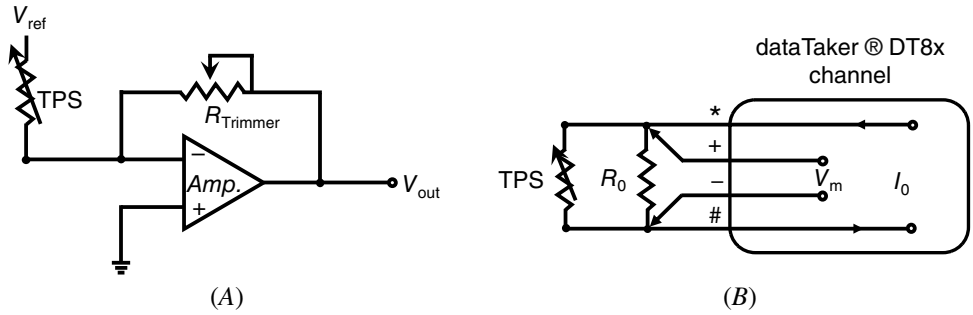


Figure 2*A* shows a method suggested by Interlink Electronics (2023) and used in some other studies (Gao, Wang, and Chow 2017). This method adds an amplifier (denoted as *Amp.* in the figure) and a trimmer resistor with a constant and known resistance R_{Trimmer}. The resultant electrical interface could convert the electrical conductance signal to a voltage signal. The electrical conductance of TPS can be calculated as follows:

$$G_{\text{TPS}} = -\frac{V_{\text{out}}}{R_{\text{Trimmer}} V_{\text{ref}}} \tag{1}$$

where V_{out} is the output electrical voltage and V_{ref} is the input reference electrical voltage. The processes of wiring and debugging using this method are troublesome and complicated. Hence, this method is suitable for cases where the number of measurement targets is limited.

Figure 2*B* is a newly proposed method that is more convenient and available for a large number of measurement targets. This method is based on a specific data logger—dataTaker DT85 manufactured by Thermo Fisher. Compared with other common data loggers in geotechnical testing, dataTaker DT85 can measure the electrical resistance (the reciprocal of electrical conductance) directly using the 4-wire Kelvin method via the corresponding module that has been embedded in the dataTaker DT85. Users can easily connect the TPSs to dataTaker DT85, as illustrated in **figure 2*B***. One problem found by the authors is that the measurement is easily affected by electromagnetic interference from electromechanical apparatuses in the experimental environment. It is because the default 4-wire Kelvin method excitation current in dataTaker DT85 for the electrical resistance range of TPS (normally 0.1∼100 kΩ) is as weak as 2 or 200 μA. To minimize the effects of electromagnetic interference, a stronger excitation current of 2.5 mA (I_0 in **figure 2*B***, optional in dataTaker DT85) is recommended. However, it would limit the maximum resistance measurement range to 700 Ω. Thus, a resistor with a resistance, R_0, of 700 Ω is suggested to be in parallel with TPS. Then, the electrical conductance of the TPS could be calculated as

$$G_{\text{TPS}} = \frac{1}{R_{\text{TPS}}} = \frac{1}{R_{\text{out}}} - \frac{1}{R_0} \tag{2}$$

where R_{out} is the output electrical resistance value. Meanwhile, common methods, such as using a grounding system and shielded twisted pair cables, should also be employed to eliminate electromagnetic interference.

DT85 also has an advantage in multichannel measurement. One set of DT85 Series 4 is available for the measurement of 16 electrical resistance signals, simultaneously. Coupled with the channel expansion module, CEM20, the maximum number of electrical resistance signal measurements is expandable to 320. In the physical modeling of this study, there are 22 TPS-based sensors, including 4 TPS-EPCs and 18 TPS-ICPTs, and thus, one set of DT85 Series 4 and one set of CEM20 are used.

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Postinstallation Calibration of TPS-EPC and TPS-ICPT

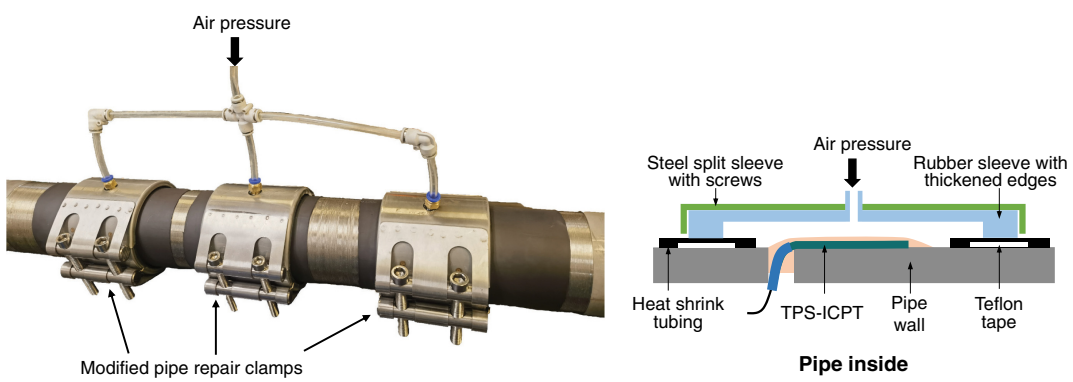
The performance of TPS highly depends on specific installation conditions. Hence, calibration after installation is necessary. A sealed tank is used for the calibration of TPS-EPCs. TPS-EPCs are put inside of the tank and its wire exits through sealed joints. During calibration, the air pressure can be controlled using a pneumatic regulator with a preset pressure path.

As for TPS-ICPTs, modified pipe repair clamps are used, as shown in **figure 3**. The pipe repair clamp, consisting of a rubber sleeve with thickened edges and a steel split sleeve with screws, was originally used for repairing leaking pipes. A through hole is punched in the middle of the repair clamp and is connected hermetically with a pneumatic connector. Hence, a sealed space that can control the air pressure is formed on the pipe surface with TPS-ICPTs (see **fig. 3*B***). Teflon tape and heat shrink tubing can be used to prevent gas leakage between the pipe and the edges of the repair clamp's rubber sleeve. Using a similar method of controlling pressure as for TPS-EPC, the relationship between applied pressure and output can be determined.

Figure 4 shows the typical results of the electrical conductance evolution over elapsed time for TPS-EPC and TPS-ICPT. The air pressure was increased stepwise from 0 to 200 kPa (loading path), a typical range in geotechnical engineering, and then reverted to 0 kPa (unloading path), in a total of 17 stages. At each stage, the pressure was swiftly adjusted and sustained for a duration of $\Delta P/v$, where ΔP represents the variation in air pressure and v signifies the loading rate, equaling 4 kPa/min. **Figures 5*A*** and **6*A*** show a detailed analysis of the results from **figure 4**, showing the relationship between electrical conductance and pressure during the loop, loading, and unloading paths, respectively. The data points for the loading and unloading paths in **figure 6*A*** represent the outputs at the end of each stage in **figure 4**, whereas the data for the loop path in **figure 5*A*** are the mean values of calibration results under the same pressure during loading and unloading paths.

Three key characteristics can be observed in **figures 4**, **5*A***, and **6*A***. Firstly, the electrical conductance of the TPS rises and falls with pressure. However, the curves representing the relationship between electrical conductance and pressure are nonlinear, with their slopes diminishing as pressure increases. Secondly, a noticeable hysteresis is present in the loop of the electrical conductance-pressure curves. Although the outputs revert to nearly initial values when the pressure is regulated to zero, the electrical conductance during the unloading path is typically higher than that during the loading path. The behavior observed during the unloading path is linked to the maximum calibration pressure. Thirdly, a short-term drift is evident in each stage, especially during the loading path. Despite the TPS's rapid response to pressure changes (less than 3 μs as stated in Interlink Electronics [2023]), the output value of electrical conductance continues to rise over time. Postinstallation calibration of 18 TPS-based sensors (comprising 15 TPS-ICPTs and 3 TPS-EPCs) regarding these features are provided below.

FIG. 3 Proposed calibration method of TPS-based ICPT: (*A*) photograph and (*B*) schematic diagram of internal details.



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FIG. 4

Typical calibration result (loading path: 0~200 kPa; unloading path: 200~0 kPa; 17 stages in total; 4 kPa/min).

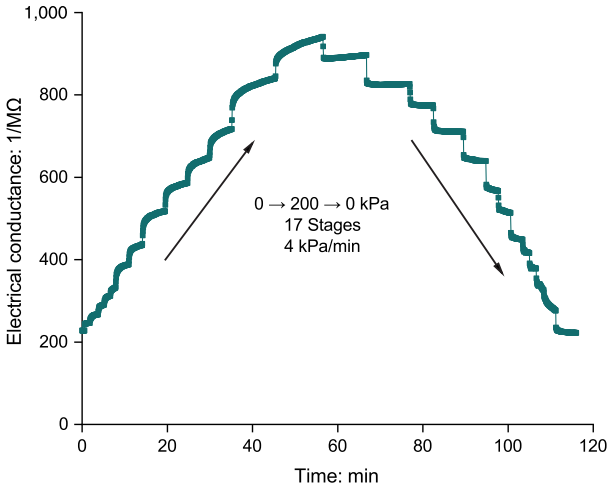
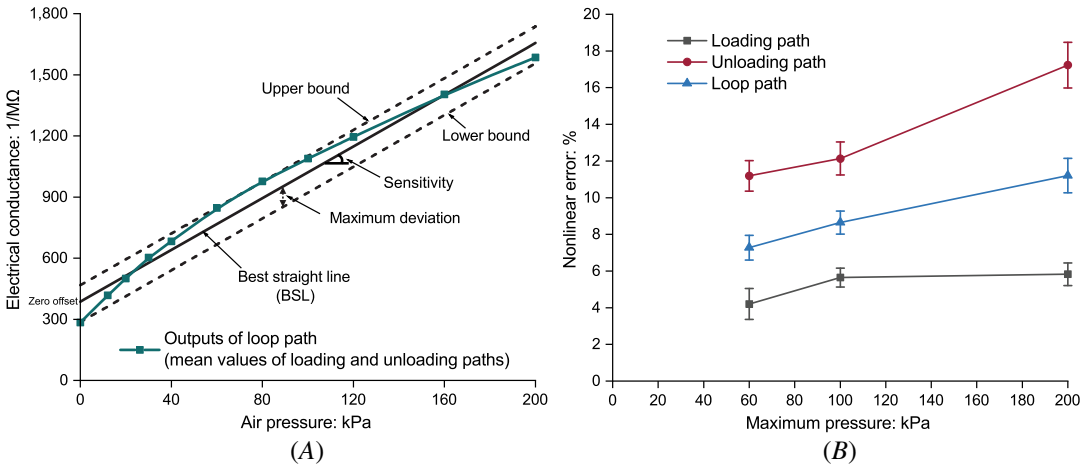


FIG. 5 Nonlinearity error of TPS-based sensor (number of TPS-ICPT = 15, number of TPS-EPC = 3, 4 kPa/min, error bar of Y-axis = ± standard error of the mean): (*A*) typical air pressure–electrical conductance relationship in loop path; (*B*) nonlinearity error versus maximum pressure.



The nonlinearity of a sensor can be gauged by its nonlinearity error. As depicted in **figure 5*A***, calibration data can be fitted using the best-fit straight line (BSL): $G = kP+G_0$, where P represents the output value of pressure, G is the measured electrical conductance, k is sensitivity, and P_0 is the zero offset of the sensor. The nonlinearity error is defined as the percentage of the maximum output deviation between the calibration curve and its BSL to the maximum output range, as outlined by Nyce (2003). **Figure 5*B*** presents the average nonlinearity error of the loading path based on the calibration results of 18 TPS-based sensors with a pressure loading rate of 4 kPa/min. The nonlinearity errors of the unloading and loop path (calculated using values of the mean of calibration results under the same pressure during loading and unloading paths) are also depicted. The nonlinearity error escalates with the maximum calibration pressure, irrespective of the path. The values of the three paths increase from 4.2 to 5.8 %, 11.2~17.2 %, and 7.3~11.2 %, respectively. The calibration results under the loading path exhibit the best linearity. The standard errors of the average nonlinearity errors are no more than ± 8.4 ‰.

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FIG. 6 Hysteresis errors of TPS-based sensor (number of TPS-ICPT = 15, number of TPS-EPC = 3, 4 kPa/min, error bar of Y-axis = ± standard error of the mean): (*A*) typical air pressure–electrical conductance relationship in loading and unloading paths; (*B*) hysteresis error versus maximum pressure.

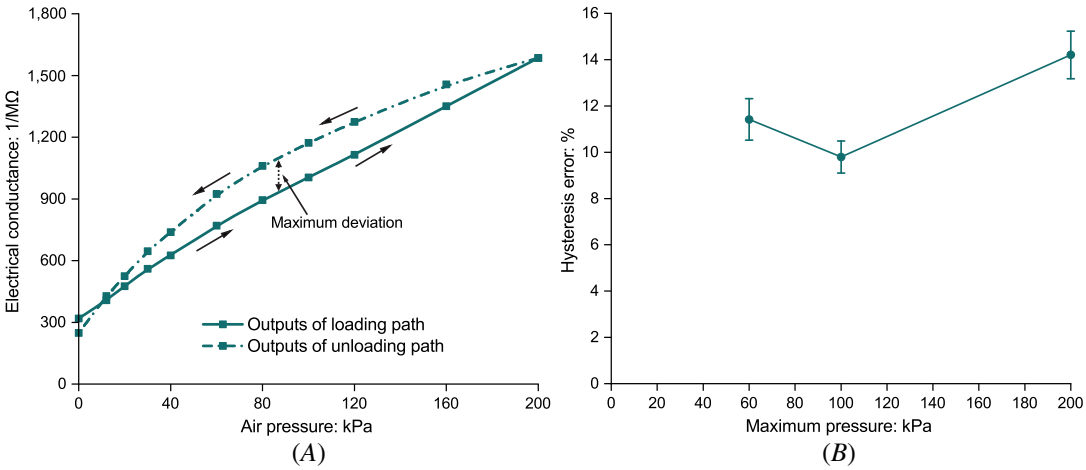


Figure 6 depicts the hysteresis errors with varying calibration pressures based on the calibration results of 18 TPS-based sensors. The hysteresis error is defined as the ratio of the maximum output deviation between the loading and unloading paths to the output range, as shown in **figure 6*A***. Although the effect of maximum pressure is not significant, the values are as high as at least 10 % (see **fig. 6*B***).

Both the nonlinearity error and hysteresis error contribute to the sum static error, which is a crucial measure of a sensor's performance. The sum static error, e_{sum}, of a sensor is normally estimated using the root-sum-of-squares method with $e_{sum} = \sqrt{\sum_{j=1}^{n} e_j^2}$, where e_j is an individual error (i.e., nonlinearity error or hysteresis error in this case) and n is the number of errors. Therefore, based on the above results, the TPS's sum static error could reach up to 22 %. However, using the nonlinear signal conversion method recommended in the following section could be helpful to improve measurement accuracy.

Figure 7 illustrates the typical calibration results of a TPS-ICPT with different loading rates. Three loading rates, 1, 4, and 10 kPa/min, which are estimated as the values in the following physical modeling, were selected. The maximum calibration pressure is 60 kPa. The electrical conductance of each stage increases with the loading rate, which is consistent with the results in **figure 4**. The calibration results of 18 TPS-based sensors indicate that as the loading rate increases from 1 to 10 kPa/min, the average sensitivity experiences a decrease from 8.61 to 7.25 $(GPa{\cdot}\Omega)^{-1}$, whereas the average zero offset decreases around 7 $M\Omega^{-1}$.

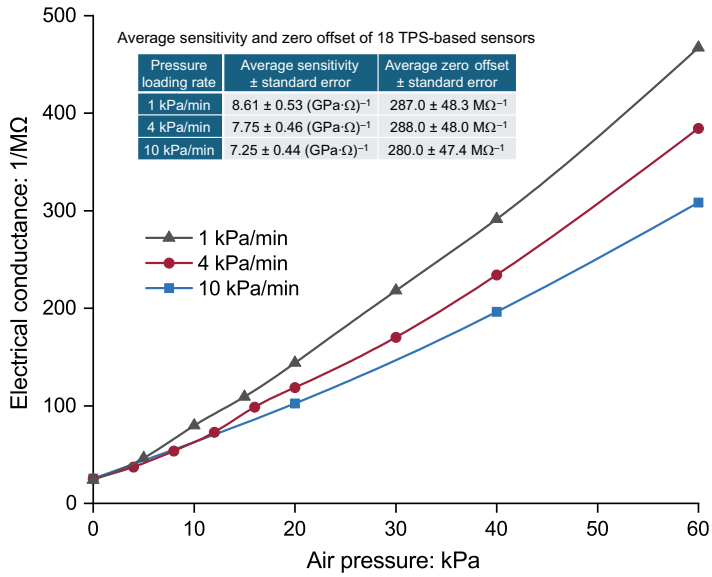
Nonlinear Signal Conversion

The relationship between the physical quantity being measured and the output signal of common geotechnical sensors, such as pore pressure transducers and linear variable displacement transducers, is normally highly linear within its measurement range. Thus, the results of the output signal are normally converted into physical quantity based on the fitted BSL. However, the performance of the TPS-based sensor is not ideal, according to the calibration results above. To improve its accuracy, a nonlinear signal conversion method is suggested to be used. After calibrating the sensor, its electrical conductance-pressure relationship is fitted using the smoothing spline algorithm in MATLAB's Curve Fitting app. Then, the results of electrical conductance measured during experiments can be input into this relationship to return the results of pressure on the sensor. Using the nonlinear signal conversion method and selecting a conductance-pressure curve calibrated under a condition close to the

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FIG. 7

Influence of loading rate on pressure-electrical conductance relationship (number of TPS-ICPT = 15, number of TPS-EPC = 3).



measurement could eliminate the effect of nonlinearity error and minimize the effect of the hysteresis error. Consequently, the deviation caused by the sum static error might be less than 7.1 %.

The loading range and rate in the calibration should be as close as possible to the experimental condition. If the measured pressure is increasing, the calibration results under the loading path should be used as the reference for signal conversion. For the unloading path, aside from the nonlinearity, the curve of the unloading path is highly dependent on the maximum calibration pressure and is only reliable when the maximum calibration pressure is close to the maximum pressure in the experiment. Conversely, if the measured pressure continually fluctuates, the calibration results of the loop path may serve as a better reference.

Comparison between Traditional EPC and TPS-Based EPC

As previously mentioned, the large thickness of traditional EPC can lead to stress concentration due to arching when they are buried inside the soil. Hence, the earth pressure measured by them is higher than the actual value. The TPS-based EPC, with a limited thickness, can significantly minimize this problem. To quantify this effect, three calibration tests were conducted using a modified oedometer, as illustrated on the right side of **figure 8**. A piezoelectric-ceramics–based EPC with a thickness (height) of 10 mm and a diameter of 20 mm represents the traditional EPC. Its nominal maximum pressure is 100 kPa with a static error of no more than 0.4 %. In condition I, the traditional EPC was embedded in the baseplate of the oedometer to create an ideal condition with no EPC thickness and compared with conditions II and III, where the traditional EPC and TPS-EPC were placed on the baseplate of the oedometer to simulate the buried conditions in the experiment and practice.

The testing soil used in this study is standard medium sand sourced from Fujian Province, China, with a particle size ranging from 0.25 to 0.5 mm. The parameters of this sand are provided in **Table 1**. The dimensions of the soil specimen inside the oedometer are 40 mm in height and 70 mm in diameter. The specimens were compacted in layers of 10 mm with a relative density of 85 %. After that, the specimen was loaded in stages. Each stage lasted at least 2 min until the readings of settlement and pressure were stable.

Figure 8 illustrates the experimental results. In condition I, the measured pressures are consistent with pressures applied on the top of the specimen, confirming the good performance of the traditional EPC when

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FIG. 8 Comparison between TPS-based EPC and traditional EPC in the oedometer.

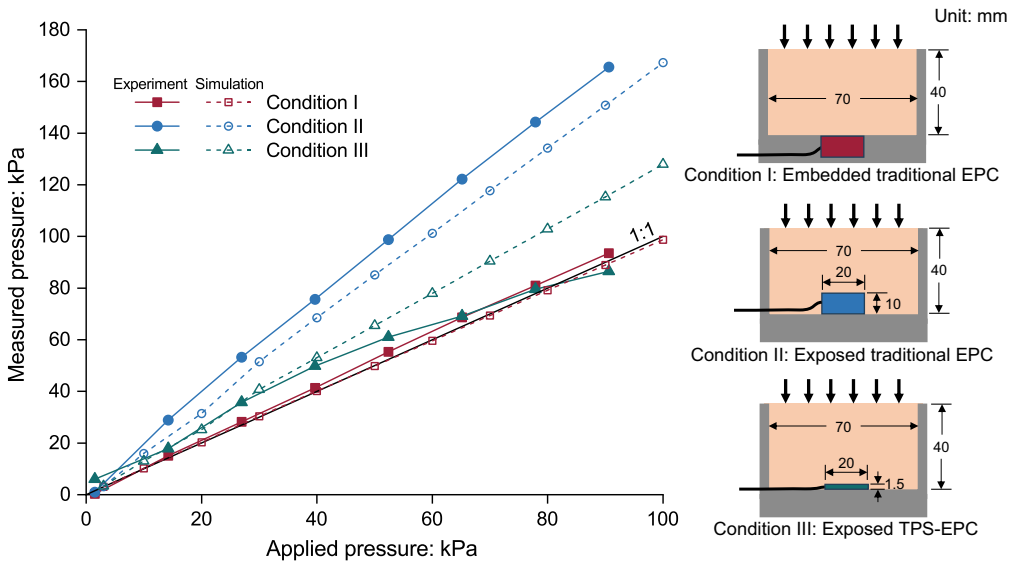


TABLE 1
Sand properties

Soil Parameter	Value
Specific gravity, G_s	2.68
Particle size, mm	0.25~0.5
Median particle size, d_{50}, mm	0.375
Maximum void ratio, e_{max}	0.797
Minimum void ratio, e_{min}	0.526
Target relative density, I_D, %	85
Dry unit weight at target relative density, γ', kN/m^3	17.0
Peak friction angle, °	39.6
Critical state friction angle, °	32.8

there is no thickness effect and piezoelectric ceramics used by the selected traditional EPC is a mature technique in pressure measurement. The measured pressures in condition II exhibit a linear relationship with applied pressures, but the former is approximately 80 % larger than the latter. This difference demonstrates the significant influence of the traditional EPC's thickness. As for condition III, the measured pressure is slightly higher than the applied values at early loading stages and then approaches the applied value at later stages. Its error is no more than 10 %, demonstrating the good performance of TPS-based sensors compared with traditional EPCs.

To understand the results of oedometer tests, three-dimensional (3D) discrete element modeling (DEM) was carried out to interpret the experimental results. The simulation was conducted using the open-source code YADE (Kozicki and Donzé 2008). A six-wall box was established as the boundary to model the oedometer cell. Another six-wall box set in the middle of the boundary bottom wall represented the sensor. The dimensions of the boundary and sensor model were the same as those of the experiments. The shapes of the boundary and the sensors were cuboid rather than cylinders for easier modeling. The linear elastic-friction model (Šmilauer et al. 2021) is used to simulate the particle-particle and particle-wall contacts, which requires three input parameters,

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TABLE 2
DEM simulation parameters

Simulation Parameter	Value
Particle density, kg/m^3	2,680
Particles Young's modulus, MPa	150
The ratio of shear to normal stiffness of particles	0.2
Particles contact friction angle, °	34
Sensor and boundary walls Young's modulus, MPa	19,000
The ratio of shear to normal stiffness of the sensor wall	0.7
Sensor walls contact friction angle, °	10
Boundary walls contact friction angle, °	0
Target relative density, I_D, %	85
Damping ratio	0.7

including Young's modulus, the ratio of shear to normal stiffness, and the contact friction angle. The model parameters used in this study are calibrated against the direct shear tests on the sand used in this experiment and summarized in **Table 2**. Approximately 30,000 particles with uniform diameters were randomly generated and were compacted to the target relative density (85 %) using the radius expansion method (Scholtès et al. 2009). The final particle diameter was around 1.88 mm, consistent with 5 times the mean particle size in the experiment. Finally, the normal pressure on the specimens was controlled using the top wall of the boundary cuboid. The specimens were also loaded by stages. Each stage continued until the top pressure reached the target, and the unbalanced force was less than 0.1 %.

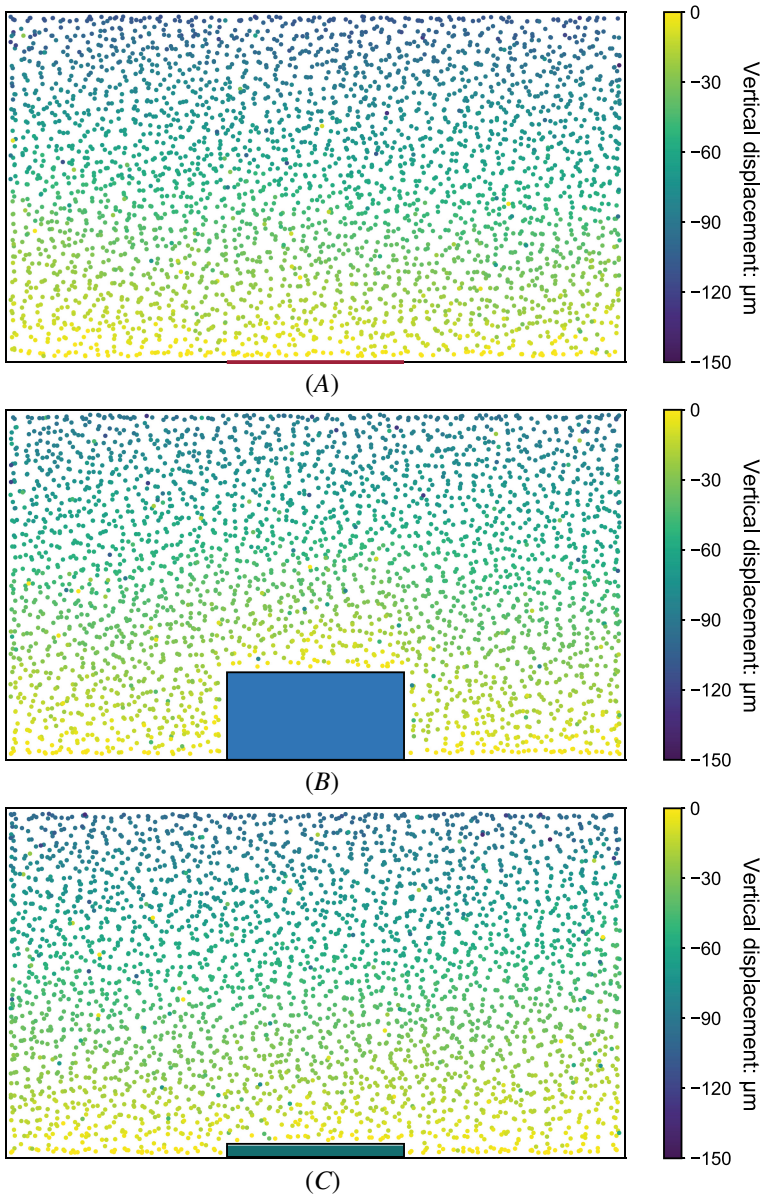
The computed pressures are also shown in **figure 8** for comparison. The trends observed in experiments are all confirmed by the numerical results, although the computed and measured results are slightly different. The arching effects mechanism can be analyzed using the DEM simulation results. **Figure 9** illustrates the vertical displacement of soil particles as the applied top pressure increases from 0 to 100 kPa. All the values are negative due to soil compaction as pressure increases. **Figure 10** shows the projection of the strong force chain network in the DEM simulation of conditions I to III. The strong force chain is a chain network of contact normal forces larger than their median values (Peters et al. 2005). The thickness of each chain is proportional to the magnitude of force. For clear distinction, only the data from particles between the sensor's front and rear walls are shown in **figures 9** and **10**. In condition I, shown in **figure 9*A***, the vertical settlements (negative values of vertical displacement) of particles decrease from around 150 μm to 0 μm with increasing depth, and all settlements are uniform at the same depth. Correspondingly, the force chains in condition I (see **fig. 10*A***) distribute uniformly in the space, with the maximum contact normal force of 2.12 N, proving that the soil particles support the loading from the top wall evenly. As for condition II, the particle settlements above the sensor's top wall are smaller than those on the left and right sides due to varying soil thickness (see **fig. 9*B***). This nonuniform settlement indicates a relative displacement, applying an additional downward force above the sensor's top wall with stronger force chains (maximum contact normal force = 3.08 N, see **fig. 10*B***). This corresponds to the higher measured pressures shown in **figure 8**. By comparison, nonuniform particle displacement and the concentration of the force chain (2.16 N in maximum) on the sensor top wall in condition III (see **figs. 9*C*** and **10*C***) is not as obvious as that of condition II, causing a limited difference between the measured and applied pressures in **figure 8**.

Based on the above tests and DEM simulations, the limited thickness of the TPS-EPC can reduce the error caused by the arching effects on the sensors from around 80 % to no more than 10 % when the sensor is buried inside the soil, demonstrating a more accurate measurement of earth pressure. Furthermore, in the practice of field or physical modeling, the soil around the buried sensors can be disturbed during sensor installation. Compared with the traditional commercial EPC, the TPS-EPC, with its smaller volume and mass, would cause fewer disturbances.

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FIG. 9

Vertical displacement with applied pressure increasing from 0 to 100 kPa in DEM simulation: (*A*) condition I; (*B*) condition II; (*C*) condition III.



Application in Physical Modelling of Axial Soil-Pipe Interaction

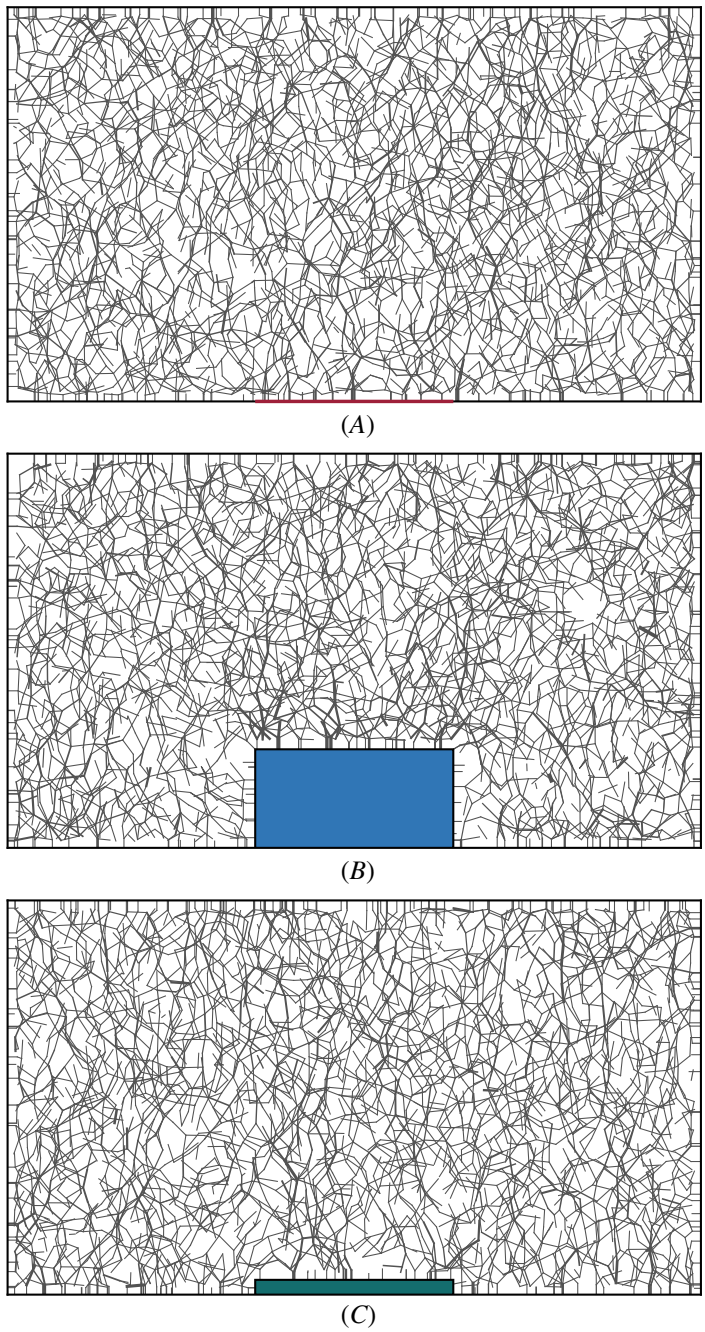
EXPERIMENTAL SYSTEM, MATERIAL, AND METHOD

The feasibility of TPS-based sensors application in physical modeling was verified in a cyclic axial soil-pipe interaction test using the apparatus in figure 11. This apparatus comprised a steel box, a steel pipe, an axial actuation subsystem, and a vertical loading subsystem. Two holes are located on the front and rear walls at a height of 0.35 m, allowing the pipe to pass through and maintain a consistent soil-pipe interaction length during the test. A seamless steel pipe, typically used in Hong Kong's natural gas and water pipeline system, with a nominal outer

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FIG. 10

Force chain network with the applied pressure of 100 kPa in DEM simulation: (*A*) condition I (the maximum contact normal force = 2.12 N); (*B*) condition II (the maximum contact normal force = 3.08 N); (*C*) condition III (the maximum contact normal force = 2.16 N).



diameter of 102 mm and a nominal thickness of 4 mm, was employed (British Standards Institution 2002; Water Supplies Department 2020). The pipe surface underwent a turning method treatment with a depth of 0.38 mm and an interval of 0.9 mm to obtain a normalized roughness ($R_n = R_{max} / d_{50}$, Kishida and Uesugi [1987]) of 1.01. Further details are summarized in **Table 3**.

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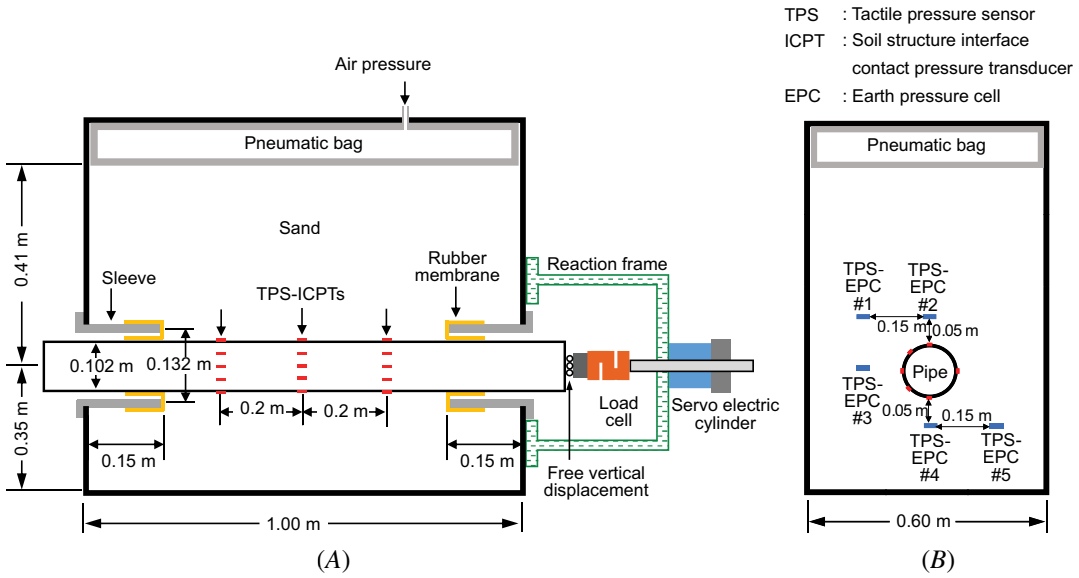
TABLE 3
Pipe properties

Pipe Parameter	Value
Diameter, D, mm	102
Thickness, mm	4
Total length, m	1.25
Testing length, m	0.7
Steel density, kg/m^3	7,930
Young's modulus, E, MPa	204
Poisson's ratio	0.3
Surface treatment method	Turning surface
Maximum height on the surface profile, R_{max}, mm	0.38
Normalized surface roughness, R_n	1.01
Peak interface friction angle, °	37.9
Critical state interface friction angle, °	33.5

The axial actuation subsystem, which comprises a servo-electric linear actuator and a reaction frame, was designed to regulate axial displacement. The pipe was engineered to move freely in the vertical direction during its axial movement, facilitated by a connector depicted in **figure 11*A***. A flexible pneumatic bag was positioned between the soil's upper surface and the top cap. The length and width of the pneumatic bag are designed to fit the interior dimensions of the box. The bag's top surface has two holes: one for connection to a pneumatic regulator and the other for an air pressure gauge. A top cap, reinforced with beams, was installed above the pneumatic bag and the box to provide a reaction force. The nominal pressure at the center, σ_c', can be manipulated by adjusting the air pressure P inside the bag, using the formula $\sigma_c' = \gamma' H_{c0} + P$, where H_{c0} (= 0.36 m) represents the actual buried depth from the ground surface to the pipe's center. The target nominal pressure σ_c' in this study is 34 kPa, equivalent to 2 m in burial depths.

Eighteen TPS-ICPTs were placed across three cross-sections of the pipe, with a spacing of 0.2 m between each section. On each cross-section, six TPS-ICPTs were installed at the crown, one shoulder, two springlines, one

FIG. 11 Experimental system schematic diagram: (*A*) front view and (*B*) side view.



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haunch, and the invert, as indicated by red points in **figure 11**. Four TPS-EPCs were employed to measure overburden pressures at various positions (see **fig. 11*B***). Four traditional EPCs, based on piezoelectric ceramics, were buried in the same distribution as the TPS-EPCs for comparison. All TPS-based sensors were calibrated under a loop path of 0–200 kPa. The loading path curves were used as a reference during the first loading (the loading process of the first cycle), as the pressures were primarily increasing. After the first loading, the loop path curves were used for conductance-pressure conversion due to repeated increases and decreases. Additionally, a load cell with a measurement range of ± 10,000 N and an linear variable differential transformer with a measurement range of 50 mm were installed between the pipe and the loading system to monitor the axial force and axial displacement, respectively.

The sand used in this study is medium sand from Fujian Province, with a particle size of 0.25 to 0.5 mm. More details of this sand are summarized in **Table 1**. The target relative density for the study is set at 85 %, as per the compaction requirement for backfilling pipeline trenches in practice (United States Department of the Interior Bureau of Reclamation 1996; Water Supplies Department 2012). The sand pluviation method was used to prepare dense and uniform samples (Rad and Tumay 1987; Fretti, Lo Presti, and Pedroni 1995; Tabaroei, Abrishami, and Hosseininia 2017).

After preparing the soil sample and placing the pneumatic bag on the soil, it was covered with the top cap. The air pressure was then set to the desired value and maintained for around 30 min until sensor readings, such as vertical displacement and earth pressure, became stable. The pipe was subjected to axial loading at a speed of 0.02 mm/s. Five one-way cycles with an amplitude of 20 mm were applied. More details of the experimental system and methodology can be seen in (Guo and Zhou 2025).

Results and Discussion

PERFORMANCE OF TPS-ICPT

Figure 12 depicts the evolution of interface contact pressures as measured by TPS-ICPTs. The discrepancy in interface contact pressures at corresponding positions across the three cross-sections is generally within 10 % of the average value. Consequently, the average values of the three cross-sections are directly displayed in **figure 12**. The difference in interface contact pressure between the left and right springlines (see **fig. 12**) remains below 4 kPa during cycling, indicating good symmetry. Therefore, the average value of the left and right springlines is used as the results of the springline in the subsequent analysis.

Before the test, the pressure at the invert exceeds 34 kPa, the nominal pressure at the center σ_c'. This could be attributed to the soil arching effects on the pipe, which is similar to the results of the arching effects on traditional EPC, as shown in **figures 8** and **9**. The stiffer pipe results in uneven deformation around the pipe, causing soil above and below the pipe to experience relative displacement to the soil on the sides of the pipe. This results in higher interface contact pressure in the vertical direction of the pipe.

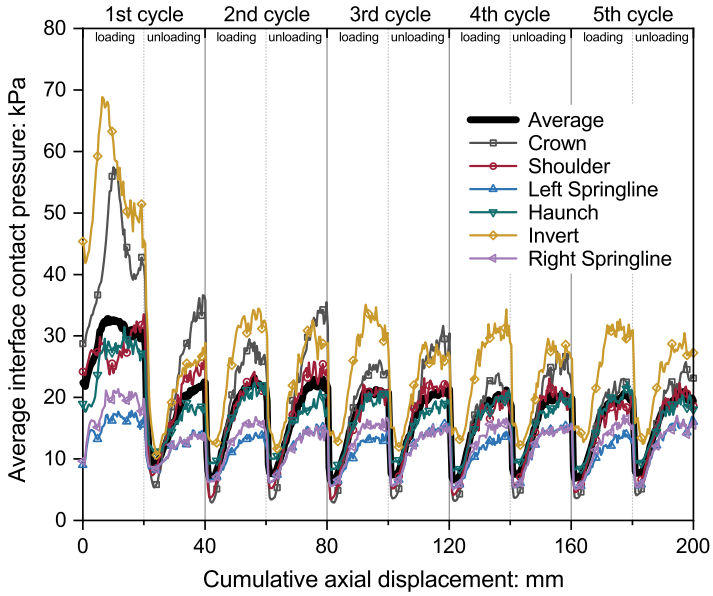
During each pullout process, there is a noticeable increase in the interface contact pressures. This is attributed to the constrained dilation behavior observed at the soil-pipe interface. Specifically, the tendency of soil dilation increases the interface contact pressures (Ng, Zhou, and Chiu 2020). The increase in pressures is particularly significant at the crown and invert, reaching 56.8 and 68.9 kPa during the first loading. The dilation between the soil-pipe interface may also cause additional relative displacement in the vertical direction, leading to further soil arching and thus an additional increase in pressure at the pipe's crown and invert. The interface contact pressures reduce to 3~12 kPa in each initial phase of the loading direction reverse, which is due to the re-establishment of the force chain.

After the first loading, the magnitude of interface contact pressures and their fluctuation range noticeably decrease. This could be due to the degradation of interface dilation during cycling and the disappearance of the arching-related strong force chain during the initial phase of the loading direction reverse. The evolution of pressures at different positions on the pipe during cycling does not follow a regular fluctuation pattern, which could be due to the inevitable misalignment of the pipe and the axial actuation subsystem. However, the average

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FIG. 12

Evolution of interface contact pressures measured by TPS-ICPTs.



interface contact pressure, as shown by the thick black curve in figure 12, displays a very regular pattern of fluctuations between 7 to 23 kPa after the first loading. The aforementioned evolution process of interface contact pressures measured by TPS-ICPT is reasonable, thereby validating the efficiency of TPS-ICPT.

The stress path at the soil-pipe interfaces, as illustrated in figure 13, further substantiates the rationality of TPS-ICPTs' results. The vertical axis of figure 13 represents the average axial resistance per unit area, calculated by the axial force measured by the load cell. Failure envelopes of steel interfaces with the same surface treatment method are determined through constant normal loading interface direct shear testing and are included as references in figure 13. The stress path shape of the first loading is similar to the typical stress path of constant normal stiffness interface shear tests by Pra-ai and Boulon (2017). It rises vertically and then turns to the top left due to the constrained dilation, as mentioned above. After touching the peak interface friction envelope, the path turns to and stops at the critical state envelope. The interface contact pressures decrease at the initial phase of the first unloading, causing the path to swiftly move to the left of the horizontal axis. Subsequently, the path gradually approaches the critical state envelope and develops to larger resistance and interface contact pressure along it. The stress paths of the following cycles have similar shapes to the first unloading and slightly move left with cycles. The evolution of the stress path illustrates that the maximum interface shear angle transforms from the peak interface shear angle to the critical state interface shear angle and finally stabilizes in the latter.

PERFORMANCE OF TPS-EPC

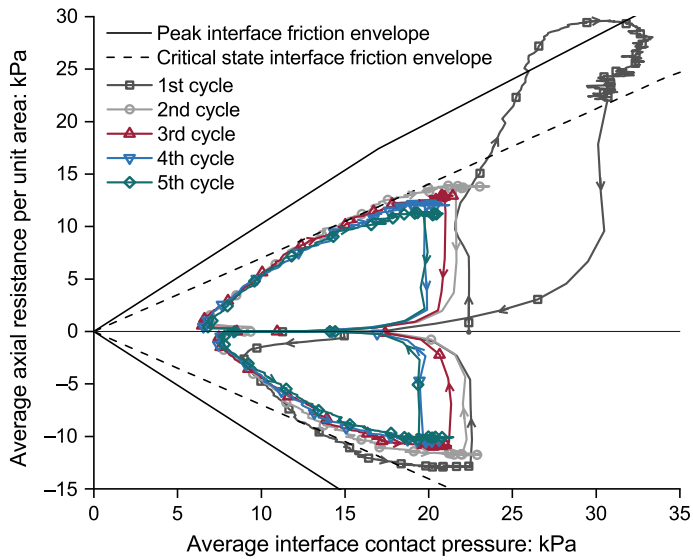
Figure 14 compares the evolution of vertical earth pressures as measured by TPS-EPCs and traditional EPCs. The overburden pressure without considering stress redistribution, σ_e', can be calculated by $\sigma_e' = \gamma' H_0 + P$, where H_0 is the buried depth from the ground surface to the sensor.

Figure 14*A* illustrates the earth pressures at a point 0.5 D higher than the pipe crown (position #1 in fig. 11*B*). Before the test, the pressure measured by TPS-EPC #1 was 35 kPa, which is slightly larger than σ_e' and the interface contact pressure at the pipe crown. As the pipe moved, the pressure increased by approximately 15 kPa due to the further arching effects and the pressure increase induced by the constrained dilation on the pipe.

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FIG. 13

Stress path based on the average interface contact pressure measured by TPS-ICPTs.



In subsequent cycles, the earth pressure increased during each loading and decreased during each reverse of the loading direction. With cycles, the measured values gradually decreased and approached σ_e', and the fluctuation amplitude decreased. This trend demonstrated the degradation of soil arching and constrained dilation. The behavior of TPS-EPC #1 is consistent with the results of TPS-ICPT and its values remain in a reasonable range. In contrast, the values measured by traditional EPC #1 reported unreliable values. Before the test, the pressure measured was 76 kPa, which is significantly larger than the estimated value. Subsequently, the pressure measured by the traditional EPC increased significantly to more than 120 kPa as the pipe continued to move. In the following cycles, the fluctuation of earth pressure measured by traditional EPC #1 remained high. From the third cycle, the earth pressure during the loading direction reverse reached 0 kPa. It might be because the continuous disturbance on the traditional EPC caused the soil around the sensor to loosen, causing no touch during the loading conversion.

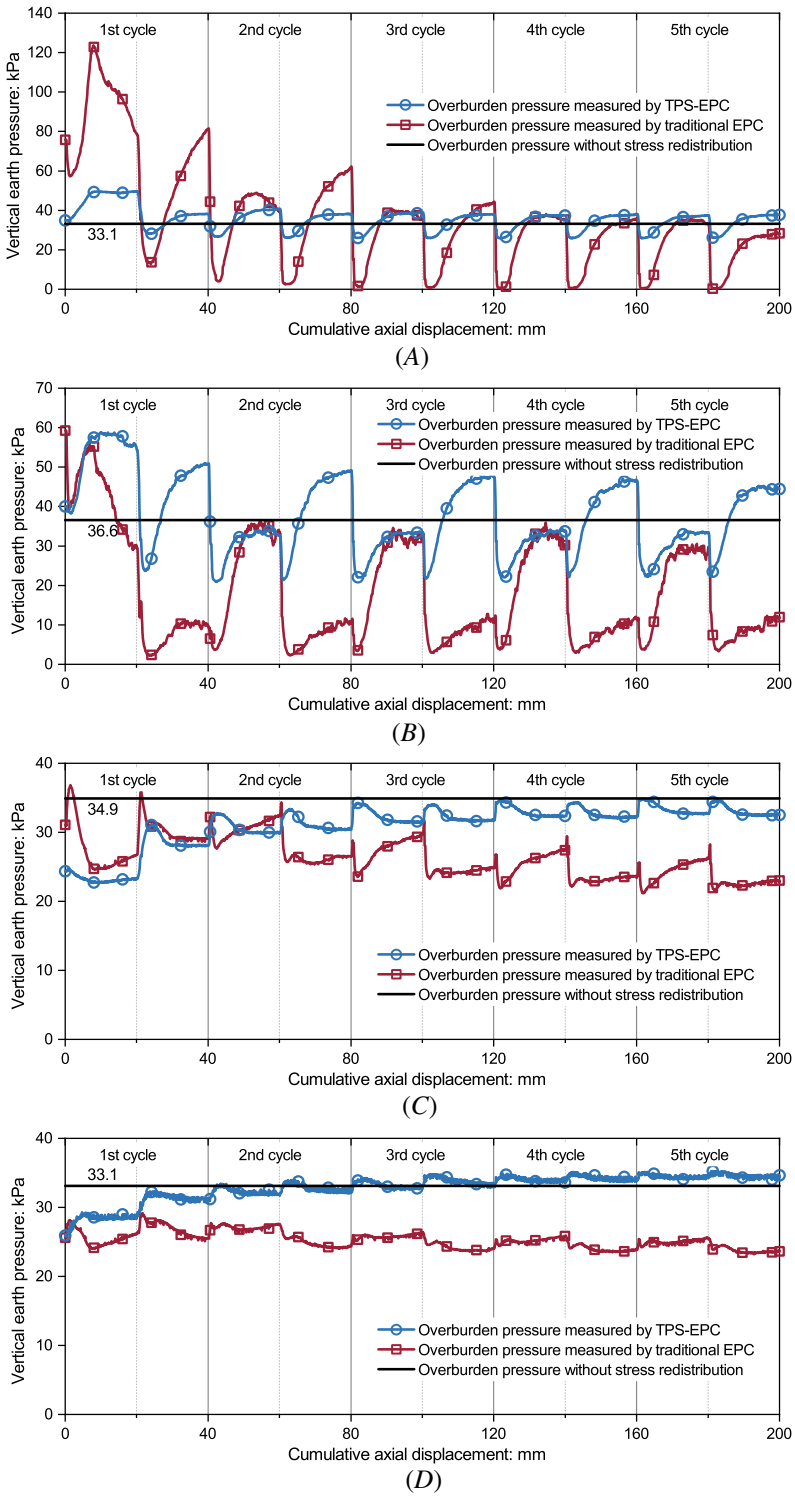
Figure 14*B* demonstrates the vertical pressure evolution at 0.5 D lower than the pipe invert (position #2 in **fig. 11*B***). TPS-EPC #2 displayed similar behavior to TPS-EPC #1, with its values fluctuating around σ_e' from the first unloading, albeit with a slightly larger amplitude. On the other hand, the pressure recorded by the traditional EPC seems unreasonable as it starts at 59 kPa, which is significantly higher than σ_e'. Subsequently, the earth pressure drops to 2.3 kPa during the next loading direction reverse, indicating excessive volatility.

Figure 14*C* and **14*D*** show the vertical pressure evolution at the position of 1.5 D to the left of the pipe center. The horizontal positions of EPCs in **figure 14*C*** and **14*D*** (#3 and #4 in **fig. 11*B***) are at the pipe center and 1.5 D higher than the pipe center, respectively. The testing results for TPS-EPC #3 and #4 exhibit contrasting behaviors to those of TPS-EPC #1 and #2. Firstly, before the test, values of TPS-EPC #3 and #4 were smaller than their σ_e'. Secondly, during the cycle, the pressures experienced decrease during each loading and unloading while increasing upon each reverse in the loading direction. Thirdly, the fluctuation amplitude gradually decreased but the outputs tended to approach σ_e'. These observations of TPS-EPC #3 and #4 further confirm the soil arching evolution, wherein the arching effects concentrate the overburden pressure on the pipe and simultaneously reduce the vertical pressure in the soil on the left and right sides of the pipe. In contrast, the outputs of traditional EPC decreased during cycles instead of approaching σ_e'. It is also potentially due to soil loosening caused by external disturbances.

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FIG. 14

Comparison between TPS-based EPC and traditional EPC in physical modeling: (*A*) top (position #1); (*B*) bottom (position #2); (*C*) left (position #3); (*D*) top left (position #4).



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Conclusions

This article introduces a new EPC (TPS-EPC) and a new soil-structure ICPT (TPS-ICPT), both based on a single-point-type TPS. The performance of TPS-based sensors is assessed and applied to a cyclic axial soil-pipe interaction physical model. The primary conclusions are as follows:

The performance of TPS is contingent on its specific installation conditions. Therefore, postinstallation calibration is crucial for TPS-based sensors. Furthermore, the sum error of TPS on nonlinearity and hysteresis could reach up to 22 % based on the BSL of calibration results. The loading rate of pressure also influences the sensitivity and zero offset of TPS-based sensors. Using the nonlinear signal conversion method and selecting a conductance-pressure curve calibrated under conditions close to the experiment can optimize accuracy.

TPS-ICPTs performed commendably in the cyclic axial soil-pipe interaction physical model. The evolution of interface contact pressures' distribution demonstrated a coupling process of soil arching effects on the pipe and the constrained dilatancy behavior on the soil-pipe interface. The average interface contact pressure exhibits a consistent pattern of fluctuations after the first loading. The stress path transitions from the peak interface shear envelope to the critical state interface shear envelope. The above proves the reliability of the results of interface contact pressures measured by TPS-ICPTs.

A comparison between TPS-EPC and traditional EPC in a modified oedometer and a corresponding 3D DEM simulation. The results reveal that traditional EPC, when buried inside the soil, is significantly affected by its thickness, leading to an 80 % overestimation of the applied pressure due to the soil arching effects on the sensor. In contrast, the limited thickness of TPS-EPC can minimize the arching effects on the sensors, resulting in a more accurate earth pressure measurement with an error of no more than 10 %. In the physical modeling, the vertical earth pressures measured by traditional EPCs fluctuated excessively. This might be due to the more pronounced arching effects on the sensor and the disturbance of soil around the sensor during pipe movement. Conversely, the values measured by TPS-EPC fluctuated around the estimated pressure and were reasonably affected by the evolution of soil arching on the pipe.

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